AMR Active Learning: Colab Setup Guide

Code: `AMR_AL/` (also the GitHub repository `amr-active-learning`). Runs on Google Colab from one Google Drive folder shared between several accounts. Every job saves progress after each round, so sessions can stop and resume in any account.

Order of work

#	Step	Who	Runtime	Time
1	Put `AMR_AL` on Drive and share it	main account	none	5 min
2	Notebook 00: setup check and T4 test	every account	T4 GPU	about 5 min
3	Dryad API account	once	none	10 min
4	Notebook 01: download DRIAMS	one account, once	CPU is enough	about 30–90 min
5	Notebook 02: audit, tuning, premise check	one account, once	T4 GPU	about 30–60 min
6	Notebook 03: experiment workers	all accounts, repeatedly	T4 GPU	see §6
7	Notebook 04: results	any account, any time	CPU is enough	a few min

1. Put the folder on Google Drive

1. In the **main account**, open drive.google.com, go to My Drive, and drag in the folder `Research/Baterial-AMR/AMR_AL`. Keep the name `AMR_AL`.
2. Right-click `AMR_AL` → Share, and add each other account as an **Editor**.
3. In **each other account**, go to "Shared with me", right-click `AMR_AL` → Organize → Add shortcut → My Drive. Do not rename it.

Storage: the compact DRIAMS-A copy is roughly 2–4 GB, and the other sites (only for E4) add more. A file counts against the account that creates it, so run notebook 01 in an account with space free. Results are only a few MB.

2. Opening a notebook, and notebook 00 (every account)

1. In Drive, open `AMR_AL/notebooks`, right-click a notebook → Open with → Google Colaboratory.
2. So that two accounts don't overwrite each other's notebook file, use **File → Save a copy in Drive** in each account and work in that copy. The copy still uses the shared folder.
3. Choose **Runtime → Change runtime type → T4 GPU → Save**. In section 4 of the notebook, set `ACCOUNT = "acc1"` (acc2, acc3 … for the others). Then **Runtime → Run all**, and allow Drive access when asked.

> **You should see:**
>
> - `compute device for XGBoost: cuda`
> - **SELFTEST: 17/17 checks passed**. The self-test runs the whole pipeline on a small synthetic dataset in a temporary folder and never touches the real data.
> - Two speed lines, for example `XGBoost fit on cuda: … s`. Note the cuda number: it sets the running time in §6.
> - The list of accounts that have written to the shared folder.
>
> If it says `cpu`, the runtime is not a T4, or that account's GPU quota is used up for today. Everything still works on CPU, just more slowly.

3. Dryad API account (once)

DRIAMS is on Dryad (doi:10.5061/dryad.bzkh1899q, free, CC0), but Dryad only allows downloads with an API token.

1. Go to datadryad.org → Log in → with ORCID. If you don't have an ORCID iD, make a free one at orcid.org.
2. Open your **My account** page and create an **API account**. It gives you a **Client ID** and a **Secret**. If the page doesn't offer this, email help@datadryad.org and ask for API access to download a public dataset.
3. In the Colab account that will run notebook 01, click the key icon ("Secrets") in the left bar and add `DRYAD_CLIENT_ID` and `DRYAD_CLIENT_SECRET`. Switch on notebook access for both.

Never paste the secret into a cell, a shared notebook or the GitHub repository. The notebook turns it into a token that lasts 10 hours.

4. Notebook 01: download DRIAMS (once, one account)

1. A CPU runtime is enough here, and it saves GPU quota. Run all.
2. **Token cell:** should print `200` and a welcome message.
3. **File list:** DRIAMS_A.tar.gz 86 GB, B 3.7 GB, C 12.4 GB, D 42.6 GB.
4. **Peek:** check that you see paths like `DRIAMS-A/binned_6000/2015/<code>.txt` and `DRIAMS-A/id/2015/2015_clean.csv`. If the folder names are different, stop and keep the printed list.
5. **DRIAMS-A:** streams for roughly 30–90 minutes, printing progress every minute (GB done, MB/s, time left, spectra kept). Keep the tab open and the laptop awake. Only the 6000-bin spectra and the label files are kept; the 86 GB archive is never stored.
6. If it stops midway, run the cell again. It restarts from the beginning, because a compressed stream can't resume.
7. DRIAMS-B, C and D are only needed for E4. Set `GET_OTHER_SITES = True` when you're ready (D alone is 43 GB to stream).

> **You should see,** for each year: the number of spectra with labels, how many label rows had no spectrum, and `has_acquisition_date` (True or False).

5. Notebook 02: audit, tuning, premise check (once, T4)

- **Audit** (`tables/audit_sites.csv`, `tables/audit_pairs.csv`): isolates and % resistant per pair and year, and the share of isolates that have acquisition dates.
- **Tuning** (`config/best_params.json`): a small XGBoost grid, 3-fold cross-validation on **2015–16 only**. Later years are never used for tuning.
- **Premise check** (`tables/premise_check.csv`): a 2015–16 model is tested on each 2017 batch and on 2018, then compared with a model retrained on all of 2017. One line per pair: `gap retraining can close = …`

> **Decide before running the workers:**
>
> 1. If `with_dates` is about 1.0, the 2017 pool arrives in real calendar quarters. If it is 0, it arrives in 4 fixed pseudo-batches; that is fine, but the paper must say so.
> 2. If a pair's gap is below about 0.02 AUROC, retraining barely matters for that pair. Discuss with the supervisors whether the cross-site question (E4) should lead.
> 3. Don't change `amr_al/config.py` after the workers start. Job names don't record model settings, so results from different settings would mix. To change something, edit the file on your Mac, drag it into Drive to replace `AMR_AL/amr_al/config.py`, and only then start the workers.

6. Notebook 03: workers (all accounts, as often as you like)

At the top of the notebook, set these four values:

Setting	Meaning
`WORKER`	A fixed name per account (`acc1`, `acc2` …). Keep it the same every time that account runs.
`SHARD`, `N_SHARDS`	With 3 accounts running at once: shards 0, 1 and 2, and `N_SHARDS = 3`. Each account does its own share first, then helps with what is left.

Setting	Meaning
`HOURS`	No new job starts after this many hours (default 3.5). This matters because free GPU sessions get cut off.
`EXPERIMENTS`	`None` runs everything in the order E1 → E3 → E2 → E4. Or name one, e.g. `["E1_temporal"]`.

- **Progress:** the status table shows done / total / running for each experiment. The log is at `logs/<WORKER>.log`.
- **Disconnected?** Run the notebook again. The job resumes from its last finished round.
- **GPU quota ran out?** Continue in another account. A job the old session had started is released after 3 hours, or immediately if you run again later with the same WORKER name.
- **Before closing a tab,** run the last cell. It flushes everything to Drive.

Experiment	Jobs	Model fits per job	GPU time if one fit takes 5 s	… if 10 s
E1_temporal	660	up to 5	≈ 4.5 h	≈ 9 h
E3_routing	180	up to 5	≈ 1.3 h	≈ 2.5 h
E2_semisup	390	up to 5	≈ 2.7 h	≈ 5.4 h
E4_crosssite	990	up to 2 (larger training set)	≈ 3–6 h	≈ 6–12 h

Estimates only. The real per-job time appears in the worker output after the first few jobs. The first fit for each pair and seed is shared across strategies, which saves about a fifth.

7. Notebook 04: results (any time, even partway through)

Output goes to the shared folder:

- `tables/E*_final.csv`: final metrics, mean and 95% CI over seeds.
- `tables/E*_labels_to_close_95pct_gap.csv`: lab tests needed to close 95% of the gap between no retraining and full retraining, with bootstrap CI.
- `tables/E*_vs_random.csv`: each strategy against random, paired over seeds (Wilcoxon, Holm-corrected).
- `figures/E*_auroc.png` and `figures/E*_vme.png`.

8. What the experiments are

ID	Question	What runs
E1_temporal	RQ1, RQ2	Train 2015–16 (20% held out for threshold and calibration). The 2017 isolates arrive in 4 batches; before each batch the model picks which ones get a lab test; retrain; test on 2018. Strategies: random, uncertainty, diversity, hybrid. Budgets of 40, 100, 200, 400 or 800 tests in total. 10 seeds. Plus *none* (never retrain) and *full* (test every isolate).
E3_routing	RQ5	Test only isolates whose conformal prediction set (90%, per class) is ambiguous: every one of them, or capped at the budget.
E2_semisup	RQ3	Confident untested isolates added as pseudo-labels ($p \geq 0.95$ or ≤ 0.05, weight 0.5): on their own, and combined with random, uncertainty or hybrid selection.
E4_crosssite	RQ4	Train on DRIAMS-A (all years); deploy at B, C and D (2018 split half pool, half test). 5 seeds.

- **Threshold:** chosen on the held-out 2015–16 isolates so that at least 90% of resistant isolates are flagged.
- **VME** (very major error): resistant reported as susceptible. **ME** (major error): the reverse.
- **op_vme:** VMEs among arriving isolates that were not tested, i.e. the harm while the model is being maintained.
- **Labels:** R and I count as resistant, S as susceptible, as in the DRIAMS authors' code.

9. Troubleshooting

Message	Fix
`AMR_AL/amr_al not found`	The shortcut is missing or was renamed (step 1.3), or you are in the wrong Google account.
`device: cpu`	Set Runtime → T4 GPU. If no T4 is available, that account's quota is used up; switch accounts or continue on CPU.
`token rejected` / HTTP 401 during download	Wrong Client ID or Secret, notebook access to Secrets is off, or the 10-hour token expired. Re-run the token cell.
`KeyError: Ceftriaxone` or species not found	A column name differs in this DRIAMS version. Check `tables/audit_pairs.csv` and correct `PAIRS` in config.py.
`FAIL <job>` in the worker output	The job is retried on the next run. If the same job keeps failing, the error text is in `logs/<WORKER>.log`.
Colab warns the notebook was changed elsewhere	Two accounts opened the same notebook file. Use File → Save a copy in Drive in each account.
Self-test not 17/17	Keep the FAIL lines; nothing else should be run until it passes.

10. Checklist for day one

1. Folder on Drive, shared, and a shortcut in every account.
2. Notebook 00 in every account shows 17/17 and a cuda speed line.
3. Dryad API account created, and the two Secrets added.
4. Notebook 01 finished for DRIAMS-A, and its summary looks sensible.
5. Notebook 02 finished. Discuss the premise-check gaps with the supervisors before starting E1.